

8. The De Moivre-Laplace theorem

Thm + (De Moivre-Laplace)

Let X be a random variable with distribution $\text{Bin}(n, p)$.

Then for every $a, b \in \mathbb{R}$:

$$\lim_{n \rightarrow \infty} P\left(a < \frac{X - np}{\sqrt{np(1-p)}} < b\right) = \Phi(b) - \Phi(a),$$

where $\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy$ is the distribution function of the standard normal distribution.

Since this theorem is about distribution functions, it's often called the global version of the result. First, we will prove the local version, which is about the density and the weight functions.

For $X \sim \text{Bin}(n, p)$ we have, heuristically,

$$P(X=k) = P(k - \frac{1}{2} \leq X \leq k + \frac{1}{2}) = P\left(\frac{k - np - \frac{1}{2}}{\sqrt{np(1-p)}} \leq \frac{X - np}{\sqrt{np(1-p)}} < \frac{k - np + \frac{1}{2}}{\sqrt{np(1-p)}}\right) \approx \phi\left(\frac{k - np}{\sqrt{np(1-p)}}\right) \frac{1}{\sqrt{np(1-p)}}$$

$\uparrow$ X only takes integer values $\uparrow$ standardization $\uparrow$ if $\frac{X - np}{\sqrt{np(1-p)}} \sim N(0,1)$

Thm 1 (De Moivre-Laplace)

Thm: (De Moivre-Laplace local form)

Let $X \sim \text{Bin}(n, p)$ with p fixed. Let A_n be nondecreasing such that $\lim_{n \rightarrow \infty} \frac{A_n}{n^{1/6}} = 0$.
Then

Then

$$\max_{k: |k-np| < \sqrt{n} A_n} \left| \frac{P(X=k) \sqrt{np(n-p)}}{\varphi\left(\frac{k-np}{\sqrt{np(n-p)}}\right)} - 1 \right| = O\left(\frac{A_n^3}{\sqrt{n}}\right).$$

Notation: big O / capital order: $f = O(g)$ if $|f| \leq C \cdot g$ for some $C \in \mathbb{R}$
Do not mistake it for the small o.

Do not mistake it for the small order, $\sigma(h^k)/h^k \xrightarrow[k \rightarrow \infty]{} 0$.

proof: Keep in mind that p is a fixed number, and k satisfies the condition of the max, that is $|k - np| < \sqrt{n} \cdot A_n$. To make the formulas more compact we will write $q_1 = (1-p)$.

$$P(X=k) = \binom{n}{k} p^k q^{n-k} = \frac{n!}{k!(n-k)!} p^k q^{n-k} = \frac{\frac{(n!)^n \sqrt{n} e^n}{(e)^k \sqrt{k} e^k \frac{(n-k)!}{e} \sqrt{n-k} e^{n-k}}}{\frac{(k!)^k \sqrt{k} e^k \frac{(n-k)!}{e} \sqrt{n-k} e^{n-k}}} \quad \text{error term } \epsilon_n =$$

Stirling's formula (actually, just the proposition we proved)

$$= \underbrace{\frac{1}{\left(\frac{n-k}{ng}\right)^{n-k}} \cdot \frac{1}{\left(\frac{k}{np}\right)^k}}_I \cdot \underbrace{\frac{1}{\sqrt{\frac{n-k}{ng}} \cdot \sqrt{\frac{k}{np}}}}_{II} \cdot \frac{1}{e d \sqrt{npq}} \varepsilon_n$$

We work on I first.

$$I = \frac{1}{\left(\frac{n-k}{nq}\right)^{n-k}} \cdot \frac{1}{\left(\frac{k}{np}\right)^k} = \frac{1}{\left(1 - \frac{k-np}{nq}\right)^{n-k}} \frac{1}{\left(1 + \frac{k-np}{np}\right)^k}$$

$$\ln(I) = -(n-k) \ln\left(1 - \frac{k-np}{nq}\right) - k \ln\left(1 + \frac{k-np}{np}\right) =$$

$$= -(n-k) \left[-\frac{k-np}{nq} - \frac{1}{2} \left(\frac{k-np}{nq}\right)^2 + O\left(\left(\frac{k-np}{nq}\right)^3\right) \right] - k \left[\frac{k-np}{np} - \frac{1}{2} \left(\frac{k-np}{np}\right)^2 + O\left(\left(\frac{k-np}{np}\right)^3\right) \right] =$$

series expansions to the second order: $\ln(1-x) = -x - \frac{1}{2}x^2 + O(x^3)$, $\ln(1+x) = x - \frac{1}{2}x^2 + O(x^3)$

$$= (k-np) \frac{(n-k)p - kq}{npq} + (k-np)^2 \frac{(n-k)p^2 + kq^2}{2n^2p^2q^2} + O\left(\left(\frac{\sqrt{n} \cdot A_n}{n}\right)^3\right) \cdot n =$$

$$= \frac{(k-np)^2}{2n^2p^2q^2} = \frac{np^2 - k(p^2 - q^2)}{2n^2p^2q^2} = \frac{np^2 - k(p-q)(p+q)}{2n^2p^2q^2}$$

$$= \frac{-2npq + np^2 - k(p-q)}{2n^2p^2q^2} (k-np)^2 + O\left(\frac{A_n^3}{\sqrt{n}}\right) =$$

$$= \frac{-npq + np(p-q) - k(p-q)}{2n^2p^2q^2} (k-np)^2 + O\left(\frac{A_n^3}{\sqrt{n}}\right) =$$

$$= -\frac{(k-np)^2}{2npq} - \frac{(p-q)(k-np)^3}{2n^2p^2q^2} + O\left(\frac{A_n^3}{\sqrt{n}}\right) = -\frac{(k-np)^2}{2npq} + O\left(\frac{A_n^3}{\sqrt{n}}\right)$$

$$= O\left(\frac{(\sqrt{n} A_n)^3}{n^2}\right)$$

$$\Rightarrow I = e^{-\frac{(k-np)^2}{2npq}} \cdot e^{O\left(\frac{A_n^3}{\sqrt{n}}\right)} = e^{-\frac{(k-np)^2}{2npq}} \left(1 + O\left(\frac{A_n^3}{\sqrt{n}}\right)\right)$$

We have a similar asymptotics for II.

$$II = \frac{1}{\left(\frac{n-k}{nq}\right)^{1/2}} \frac{1}{\left(\frac{k}{np}\right)^{1/2}} = \frac{1}{\left(1 - \frac{k-np}{nq}\right)^{1/2}} \frac{1}{\left(1 + \frac{k-np}{np}\right)^{1/2}}$$

$$\ln(II) = -\frac{1}{2} \ln\left(1 - \frac{k-np}{nq}\right) - \frac{1}{2} \ln\left(1 + \frac{k-np}{np}\right) =$$

We only look at the constant part of the same series expansions now, with first order error bound.

$$= -\frac{1}{2} O\left(\frac{k-np}{nq}\right) - \frac{1}{2} O\left(\frac{k-np}{np}\right) = O\left(\frac{\sqrt{n} \cdot A_n}{n}\right) = O\left(\frac{A_n}{\sqrt{n}}\right).$$

$$\Rightarrow II = e^{O\left(\frac{A_n}{\sqrt{n}}\right)} = 1 + O\left(\frac{A_n}{\sqrt{n}}\right)$$

Combining the details

$$P(X=k) = e^{-\frac{(k-np)^2}{2npq}} \left(1 + O\left(\frac{A_n^3}{\sqrt{n}}\right)\right) \left(1 + O\left(\frac{A_n}{\sqrt{n}}\right)\right) \frac{1}{e^{\sqrt{npq}}}$$

$\tilde{\varepsilon}_n = e^{O\left(\frac{1}{n}\right)} = 1 + O\left(\frac{1}{n}\right)$ according to Stirling's formula

$$= \frac{1}{\sqrt{npq}} \underbrace{\frac{1}{e}}_{q_d^2} e^{-\frac{(k-np)^2}{2npq}} \left(1 + O\left(\frac{A_n^3}{\sqrt{n}}\right)\right)$$

□

Observe, that in the end of the proof we used $\varphi_d(x) = \frac{1}{e^d} e^{-\frac{x^2}{2}}$ in the place of $\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$, since we don't know the value of d just yet.

Similarly, let $\Phi_d(x) := \int_{-\infty}^x \frac{1}{e^d} e^{-\frac{y^2}{2}} dy$.

proof of De Moivre-Laplace theorem:

$$\left| P\left(a \leq \frac{X-np}{\sqrt{npq}} < b\right) - (\Phi_d(b) - \Phi_d(a)) \right| \leq \left| \sum_{k=\lceil a\sqrt{npq} \rceil}^{\lceil b\sqrt{npq} \rceil} P(X=k) - \int_a^b \varphi_d(x) dx \right| \leq$$

$$\leq \left| \sum_{k=\lceil a\sqrt{npq} \rceil}^{\lceil b\sqrt{npq} \rceil} \left(P(X=k) - \frac{1}{\sqrt{npq}} \varphi_d\left(\frac{k-np}{\sqrt{npq}}\right) \right) \right| + \left| \sum_{k=\lceil a\sqrt{npq} \rceil}^{\lceil b\sqrt{npq} \rceil} \frac{1}{\sqrt{npq}} \varphi_d\left(\frac{k-np}{\sqrt{npq}}\right) - \int_a^b \varphi_d(x) dx \right|$$

triangular inequality

$n \rightarrow \infty$ it's a Riemann approximating sum for the integral, and $\varphi_d(x)$ is integrable.

Now we show that the first term also converges to zero, using the local limit theorem.

Let $c > \max(b\sqrt{pq}, a\sqrt{pq})$.

$$\max_{k: |k-np| < c\sqrt{n}} \left| P(X=k) - \frac{1}{\sqrt{npq}} \varphi_d\left(\frac{k-np}{\sqrt{npq}}\right) \right| =$$

$$= \max_{k: |k-np| < c\sqrt{n}} \left| \frac{P(X=k) \sqrt{npq}}{\varphi_d\left(\frac{k-np}{\sqrt{npq}}\right)} - 1 \right| \frac{1}{\sqrt{npq}} \varphi_d\left(\frac{k-np}{\sqrt{npq}}\right) \leq \underbrace{O\left(\frac{1}{\sqrt{n}}\right)}_{\text{bounded}} \frac{1}{\sqrt{npq}} = O\left(\frac{1}{n}\right)$$

De Moivre-Laplace local form for $A_n \equiv c$

This upper bound holds for all k 's in the sum, since

$$\lceil a\sqrt{npq} \rceil \leq k \leq \lceil b\sqrt{npq} \rceil \Leftrightarrow |k-np| < c\sqrt{n}.$$

There are only $O(\sqrt{n})$ many summands, thus the sum of the error bounds is still in the order of $O(1/\sqrt{n})$.

$\Rightarrow$ it goes to zero as $n \rightarrow \infty$. $\square$

Corollary: The constant term in Stirling's formula is $\sqrt{2\pi}$.
That is $e^d = \sqrt{2\pi}$.

proof:

$$1 = \lim_{a \rightarrow -\infty} \lim_{b \rightarrow \infty} \lim_{n \rightarrow \infty} P\left(a \leq \frac{X-np}{\sqrt{npq}} < b\right) \stackrel{\text{De Moivre-Laplace}}{=} \Phi_d(b) - \Phi_d(a) = \int_{-\infty}^{\infty} \frac{1}{e^d} e^{-\frac{x^2}{2}} dx = \frac{\sqrt{2\pi}}{e^d}.$$

$\square$

Example: Flipping a coin 40 times, what is the probability that we get exactly 20 heads.

X = number of heads

$$X \sim \text{Bin}(40, 1/2) \Rightarrow P(X=20) = \binom{40}{20} \left(\frac{1}{2}\right)^{40} \approx \underline{\underline{0,1254}}.$$

Or we can use DeMoivre-Laplace Theorem:

$$E[X] = n \cdot p = 20, \quad D[X] = \sqrt{n p q} = \sqrt{40 \cdot \frac{1}{2} \cdot \frac{1}{2}} = \sqrt{10}$$

$$\begin{aligned} P(X=20) &\stackrel{\substack{\uparrow \\ \text{continuity correction}}}{=} P(19,5 \leq X < 20,5) = P\left(-\frac{0,5}{\sqrt{10}} \leq \frac{X-20}{\sqrt{10}} < \frac{0,5}{\sqrt{10}}\right) \approx \\ &\approx \Phi\left(\frac{0,5}{\sqrt{10}}\right) - \Phi\left(-\frac{0,5}{\sqrt{10}}\right) = 2\Phi\left(\frac{0,5}{\sqrt{10}}\right) - 1 \approx \underline{\underline{0,1256}} \end{aligned}$$

Example: 400 students enrolled in a given course. They attend each class with probability 0,6 only, independently of each other. How big should the auditorium be to guarantee that everyone can sit down with probability 99%?

X = attending students

$$E[X] = 400 \cdot 0,6 = 240, \quad D[X] = \sqrt{400 \cdot 0,6 \cdot 0,4} \approx 9,8$$

k = number of seats in the classroom

$$0,99 = P(X < k) = P\left(\frac{X-240}{9,8} < \frac{k-240}{9,8}\right) = \Phi\left(\frac{k-240}{9,8}\right)$$

$$\Phi(2,32) = 0,99 \text{ and } \Phi \text{ is monotonically increasing}$$

$$\Rightarrow 2,32 < \frac{k-240}{9,8} \Rightarrow \text{We need } \underline{\underline{263}} \text{ seats!}$$